Practice workbook · Make alginate beads and give uses and advantages of immobilised enzymes

Enzymes fixed in alginate beads for lactose-free milk or sugar conversion: preparation, the enzyme named and why immobilising helps.

7 question parts: 1 worked-example retry and 6 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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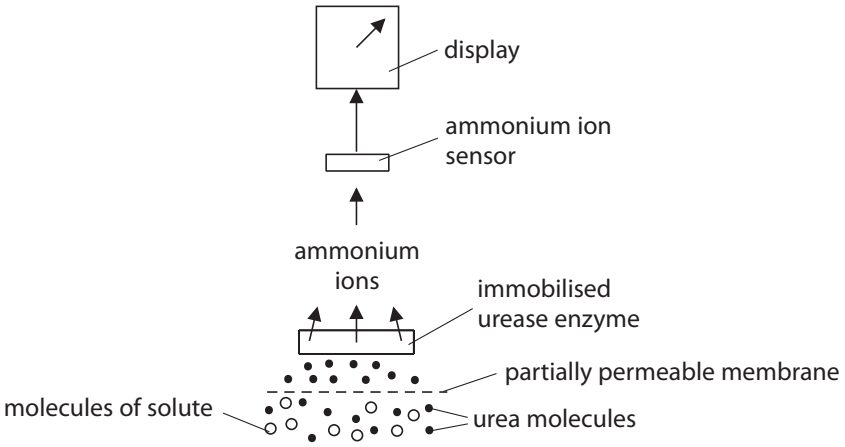
Type ID: HBI-IMMOBILISED-ENZYMES

1. Question 3 · part c · 2 marks

June 2021 Paper 2

Worked-example retry · Attempt part c only.

3 The diagram shows a biosensor that can be used to measure the concentration of urea in urine.



(a) (i) The urea molecules are in solution.

State the name of the solvent. **(1)**

(ii) Explain the function of the partially-permeable membrane. **(2)**

(iii) Suggest the function of the urease enzyme. **(3)**

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1. Question 3 · part c · 2 marks

June 2021 Paper 2

Worked-example retry · Attempt part c only.

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(b) A scientist wants to use the biosensor to compare the urea concentration of two urine samples.

Explain why it is important to do the tests at the same temperature.

(4)

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(c) Immobilised enzymes are also used in industry to produce useful molecules.

Give two advantages of using immobilised enzymes instead of enzymes that are free to move in solution.

(2)

1 .....

2 .....

(Total for Question 3 = 12 marks)

1. Question 3 · part c · 2 marks · Working and answer

Working and answer

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2. Question 6 · part a.i · 1 mark

November 2020 Paper 1

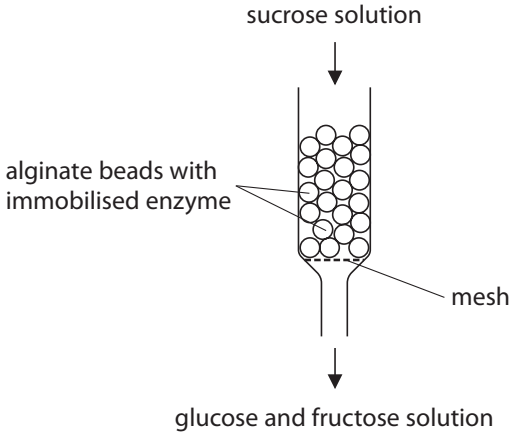
Further practice · Attempt part a.i only.

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6 (a) The diagram shows how immobilised enzymes can be used to convert sucrose into glucose and fructose.



(i) Name the immobilised enzyme used in this process.

(1)

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(ii) One advantage of using immobilised enzymes is that they do not contaminate the products, glucose and fructose.

Explain how the products could be tested to show that they had not been contaminated by the enzyme.

(3)

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(iii) State two other advantages of using immobilised enzymes.

(2)

1 .....

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2 .....

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2. Question 6 · part a.i · 1 mark · Working and answer

Working and answer

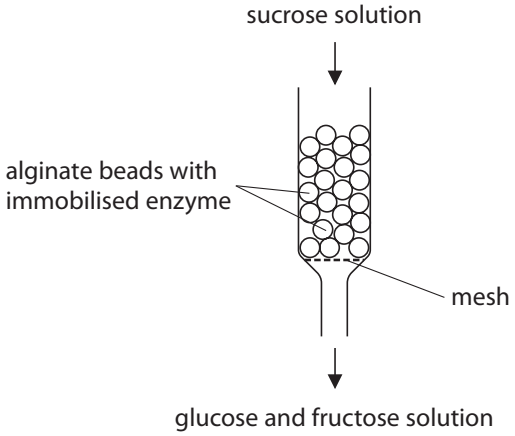
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3. Question 6 · part a.iii · 2 marks

November 2020 Paper 1

Further practice · Attempt part a.iii only.

6 (a) The diagram shows how immobilised enzymes can be used to convert sucrose into glucose and fructose.



(i) Name the immobilised enzyme used in this process.

(1)

.....

(ii) One advantage of using immobilised enzymes is that they do not contaminate the products, glucose and fructose.

Explain how the products could be tested to show that they had not been contaminated by the enzyme.

(3)

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(iii) State two other advantages of using immobilised enzymes.

(2)

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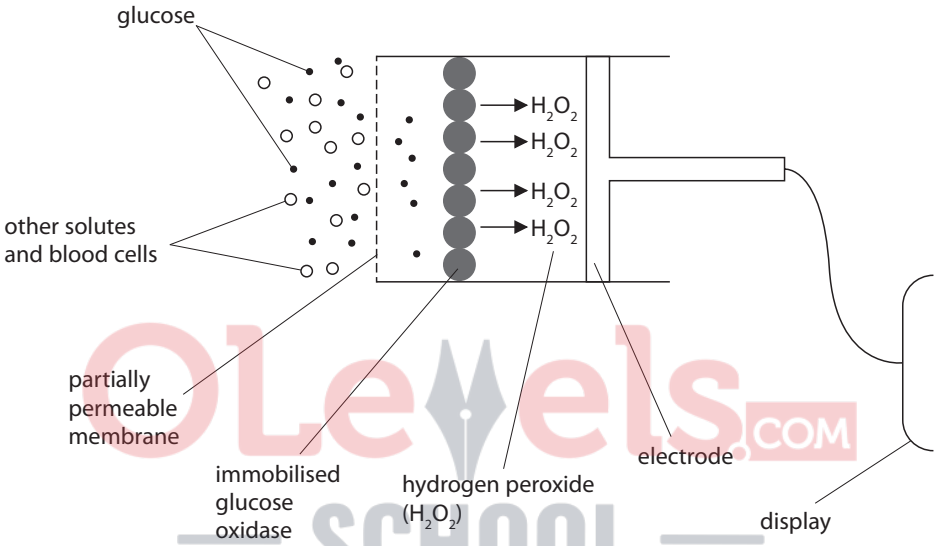
3. Question 6 · part a.iii · 2 marks

November 2020 Paper 1

Further practice · Attempt part a.iii only.

(b) Biosensors that contain immobilised enzymes can be used to measure the levels of glucose in blood.

The diagram shows this type of biosensor.



(i) Explain the function of the partially permeable membrane.

(2)

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3. Question 6 · part a.iii · 2 marks · Working and answer

Working and answer

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4. Question 7 · part a.i · 1 mark

June 2022 Paper 1

Further practice · Attempt part a.i only.

7 (a) Industrial processes can use immobilised enzymes.

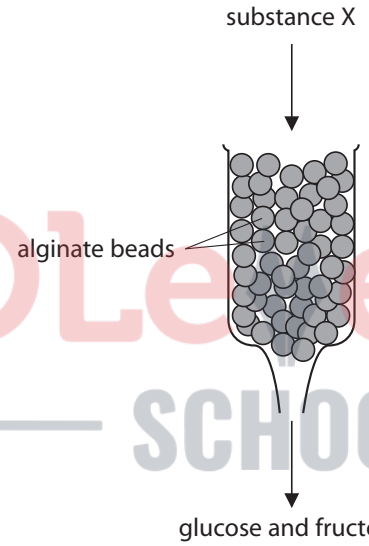
(i) Give one reason why using immobilised enzymes can cost less money than using enzymes that are not immobilised.

(1)

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.....

(ii) The diagram shows the apparatus used to produce glucose and fructose using immobilised enzymes attached to alginate beads.



Name substance X.

(1)

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(iii) Describe how to prepare alginate beads.

(3)

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4. Question 7 · part a.i · 1 mark · Working and answer

Working and answer

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5. Question 7 · part a.ii · 1 mark

June 2022 Paper 1

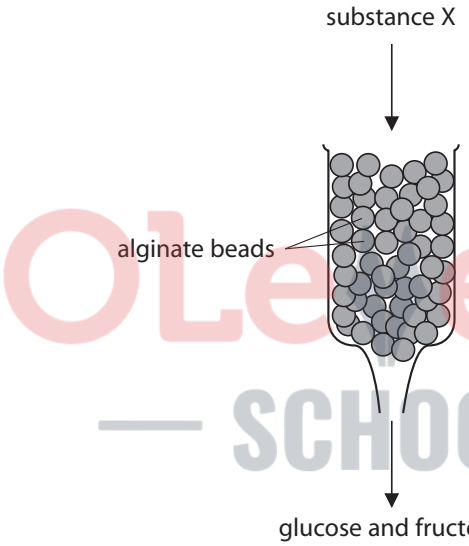
Further practice · Attempt part a.ii only.

7 (a) Industrial processes can use immobilised enzymes.

(i) Give one reason why using immobilised enzymes can cost less money than using enzymes that are not immobilised.

(1)

(ii) The diagram shows the apparatus used to produce glucose and fructose using immobilised enzymes attached to alginate beads.



Name substance X.

(1)

(iii) Describe how to prepare alginate beads.

(3)

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5. Question 7 · part a.ii · 1 mark · Working and answer

Working and answer

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6. Question 6 · part c.i · 2 marks

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Further practice · Attempt part c.i only.

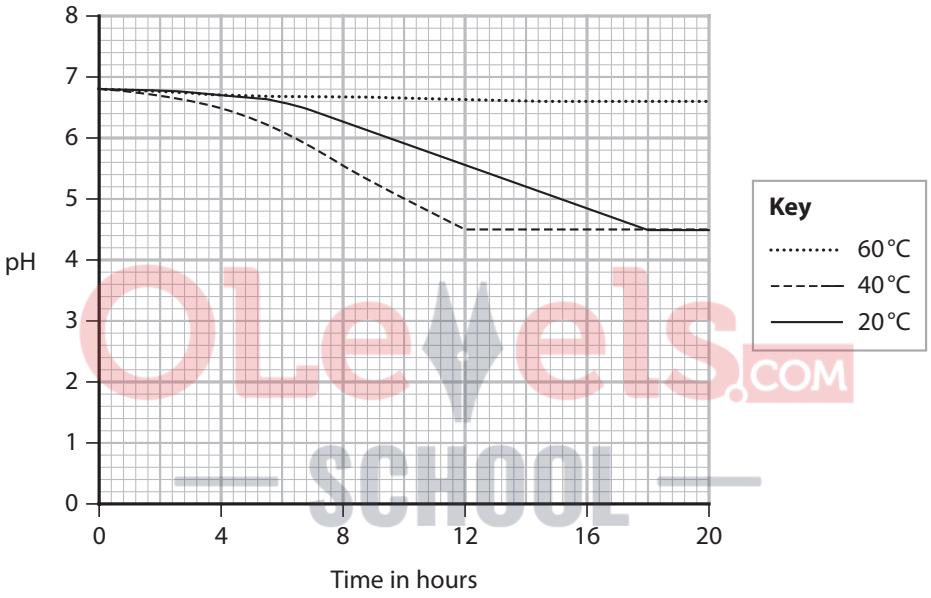
6 Yoghurt can be produced by incubating milk with a small amount of live yoghurt.

The live yoghurt contains *Lactobacillus* bacteria.

Enzymes convert lactose in the milk into lactic acid causing the pH to decrease to around pH 4.4.

In an investigation, the change in pH as yoghurt is produced was measured during 20 hours at three different temperatures.

The graph shows the results.



(a) (i) Calculate the difference in the time taken to produce yoghurt at 40 °C instead of 20 °C.

(2)

time = ... hours

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6. Question 6 · part c.i · 2 marks

January 2023 Paper 1

Further practice · Attempt part c.i only.

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(ii) Give two variables that need to be controlled in this investigation.

(2)

1 .....

.....

2 .....

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(iii) Describe how this investigation could be improved to find the optimum temperature for the production of yoghurt.

(2)

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(iv) In some methods to produce yoghurt, the milk is boiled and then cooled before the bacteria are added.

Give the reason for boiling and the reason for cooling the milk.

(2)

boiling

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cooling

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6. Question 6 · part c.i · 2 marks

January 2023 Paper 1

Further practice · Attempt part c.i only.

(b) When yoghurt reaches a pH of 4.4 the bacteria stop fermenting lactose.

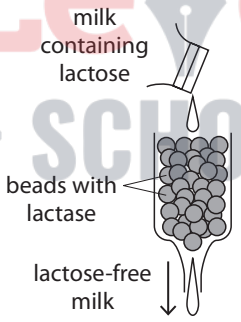
Explain why bacteria stop fermenting the lactose at a pH of around 4.4

(2)

(c) People who are lactose intolerant can drink lactose-free milk.

This can be produced using alginate beads containing lactase.

The diagram shows the equipment needed.



(i) Describe the advantages of using alginate beads containing lactase rather than a solution of lactase.

(2)

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6. Question 6 · part c.i · 2 marks

January 2023 Paper 1

Further practice · Attempt part c.i only.

(ii) The enzyme lactase breaks down lactose into simpler sugars.

Explain how the structure of the small intestine allows the absorption of these sugars.

(3)

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(Total for Question 6 = 15 marks)

Working and answer

7. Question 7 · part a.iii · 3 marks

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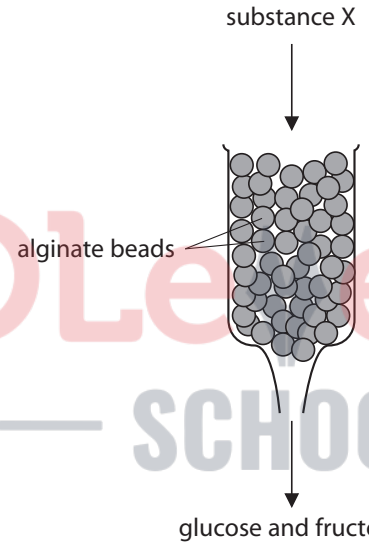
Further practice · Attempt part a.iii only.

7 (a) Industrial processes can use immobilised enzymes.

(i) Give one reason why using immobilised enzymes can cost less money than using enzymes that are not immobilised.

(1)

(ii) The diagram shows the apparatus used to produce glucose and fructose using immobilised enzymes attached to alginate beads.



Name substance X.

(1)

(iii) Describe how to prepare alginate beads.

(3)

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7. Question 7 · part a.iii · 3 marks

June 2022 Paper 1

Further practice · Attempt part a.iii only.

(b) Which process is involved in the conversion of glucose into glycogen in the human body?

(1)

- ☒ **A** glucagon is released from the pancreas and travels to the liver
- ☒ **B** insulin is released from the pancreas and travels to the liver
- ☒ **C** glucagon is released from the liver and travels to the pancreas
- ☒ **D** insulin is released from the liver and travels to the pancreas

(c) (i) The total population of the United Kingdom is 66.65 million.

Diabetes affects 6.0% of this population.

Calculate the number of people with diabetes in the United Kingdom.

(2)

number of people = ... million

(ii) Surveys were carried out in different areas of the United Kingdom.

These surveys collected data on the number of people in each area with diabetes.

The information collected is shown in the box.

38	62	51	96	49	78	82	91	65

Determine the median number of people with diabetes.

(2)

median = ...

(Total for Question 7 = 10 marks)

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7. Question 7 · part a.iii · 3 marks · Working and answer

Working and answer

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